

TO YIN YU

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EDUCATION

University of Maryland, College Park — College Park, MD

Bachelor of Science, Computer Science · December 2022

Relevant coursework: Advanced Data Structures, Artificial Intelligence, Data Science, Computer Architecture, Object-Oriented Programming, Functional Programming, Web Development, Computer Systems, Programming Handheld Systems.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, Swift, C, Rust, OCaml, Ruby, SQL

ML & Data: TensorFlow, Keras, scikit-learn, Pandas, NumPy, matplotlib, seaborn, SAS, Minitab

Web & Backend: React, Next.js, Astro, FastAPI, Tailwind CSS, PostgreSQL, Redis

Tools: Docker, Git, Linux/Unix, Vercel, Netlify, Jupyter

PROJECTS

all-in-one-URL — Short URLs, QR Codes & Barcodes | Python, FastAPI, PostgreSQL, Redis, Docker, PyJWT, bcrypt, React 19, TypeScript, Vite, Tailwind CSS 4

all-in-one-url.vercel.app

- Built a unified platform for short URLs, QR codes, and barcodes with per-resource click/scan analytics and BeautifulSoup-based automatic page-title extraction.
- Implemented SlowAPI tiered rate limiting (10 / 60 / unlimited req/min) and Redis-backed response caching plus counter batching to reduce hot-path database writes.
- Added optional JWT-based authentication (HS256 PyJWT + bcrypt-hashed passwords) with a non-destructive nullable user_id FK on existing resources, fail-fast startup checks for missing secrets, and a React Context + portaled modal auth UI.
- Containerized backend services with Docker Compose for local development, then deployed to Render (web service) with Neon (serverless Postgres) and Upstash (serverless Redis).
- Shipped a tabbed React 19 + Vite + TypeScript + Tailwind CSS 4 frontend with TanStack Query and shadcn-style components, deployed on Vercel.

United Front Roofing — Marketing Site & AI Voice Agent | Astro, TypeScript, Tailwind CSS, ElevenLabs Conversational AI, Cal.com API, Node.js, Netlify

united-front-roofing.netlify.app

- Designed and shipped a complete Astro + Tailwind marketing site for a local roofing business — landing pages, project gallery, SEO routing, a rule-based text chat assistant, and a multi-step quote estimator.
- Built an AI voice agent on ElevenLabs Conversational AI that answers FAQs, triages emergency calls (active-leak routing), qualifies leads, and books inspections end-to-end; configured webhook tools that call the Cal.com v2 API for real-time slot availability and authenticated booking creation directly from the live conversation.
- Versioned the agent's system prompt and knowledge base as markdown with a repeatable Node setup script that injects the current date into the prompt and patches the live agent in seconds — diagnosed a stale-year tool-call bug by inspecting ElevenLabs conversation transcripts via the API.
- Embedded the agent widget site-wide through a single Astro layout, coexisting with the existing text chatbot; locked the agent to the production domain via ElevenLabs allowlist.

Personal Portfolio | Next.js 16, TypeScript, Tailwind CSS v4, Framer Motion

toyinyu.vercel.app

- Built a responsive portfolio with dark/light theming, scroll-driven animations, a bento-grid project layout, and cursor-tracked spotlight cards.
- Implemented an in-house, React-19-clean theme provider with FOUC-prevention via a *beforeInteractive* inline script and a working accessible Contact form backed by Web3Forms.

Obesity Data Analysis | Python, Pandas, NumPy, scikit-learn, matplotlib, seaborn

tonyx1998.github.io

- Analyzed obesity patterns from public-health datasets, surfacing demographic and lifestyle trends with matplotlib/seaborn visualizations.
- Built and validated predictive models with scikit-learn and statsmodels to identify statistically significant risk factors.